

Research Proposals: Federated Learning

Three proposals, each revised to resolve the structural issues identified during evaluation. Dataset sources, pros/cons, and practical considerations are added below for each.

Proposal 1: Federated Fraud Detection Under Real Institutional Heterogeneity

Abstract

Federated learning algorithms for fraud detection are almost universally validated on synthetic non-IID partitions of a single dataset — one dataset split unevenly by label or via a Dirichlet distribution to approximate heterogeneity. Real institutions differ far more messily: different schemas, currencies, fraud base rates, customer populations, and data quality. This proposal builds a federation from genuinely distinct fraud datasets rather than partitions of one, and tests whether federated algorithm rankings reported under synthetic heterogeneity survive contact with real cross-institutional heterogeneity.

1. Problem Statement

It is currently unknown whether FL robustness results — SCAFFOLD or FedProx outperforming FedAvg under label skew, for instance — generalize to the heterogeneity real financial institutions actually exhibit. This is the exact gap between "FL works in the lab" and "FL works when a bank tries it."

2. Related Work and Positioning

Prior FL-for-fraud work (FedAvg/FedProx/SCAFFOLD benchmarking) is validated on Dirichlet-partitioned single datasets. The closest precedent for this proposal's format is FLamby, a cross-silo healthcare FL benchmark built from genuinely distinct hospital datasets rather than synthetic splits. This proposal is the fraud-detection analog, with an added contribution: a direct, controlled comparison between synthetic and real heterogeneity on the *same* algorithm set, rather than only reporting one setting.

3. Dataset Sources

Dataset	Source	Access	Role
IEEE-CIS Fraud Detection	Kaggle competition, released by Vesta Corporation — https://www.kaggle.com/c/ieee-fraud-detection	Public, free download, Kaggle competition rules apply to redistribution	Institution 1; also internally Dirichlet-partitioned for the synthetic-heterogeneity control
PaySim	Kaggle, originally introduced in Lopez-Rojas et al. as a mobile-money fraud simulator — https://www.kaggle.com/datasets/ealaxi/paysim1	Public, free download, CC-BY-SA style license	Institution 2
ULB Credit Card Fraud	Machine Learning Group, Université Libre de Bruxelles — https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud	Public, free download, ODbL license	Minimum-common-schema stress test only (excluded from primary comparison)

All three are publicly downloadable without a data use agreement, which keeps the project executable without an institutional review process — a meaningful practical advantage over Proposal 2.

4. Proposed Methodology

Federation construction. IEEE-CIS Fraud Detection and PaySim serve as two genuinely distinct "institutions." ULB Credit Card Fraud is excluded from the primary comparison — its features are PCA-anonymized components with no cross-dataset semantic meaning, so including it would force the shared feature set down to transaction amount and time only, making any ranking result inconclusive. ULB is retained as a separate, explicitly labeled *minimum-common-schema stress test* rather than blended into the main result.

A synthetic-heterogeneity control federation is built by Dirichlet-partitioning IEEE-CIS internally by a real-world proxy variable (merchant category or device type), giving a direct within-study synthetic vs. real comparison on the same base data.

Schema harmonization. A shared minimal feature set (amount, time, a small number of engineered aggregate features such as transaction velocity) is engineered across IEEE-CIS and PaySim. This alignment step is documented as a secondary contribution, since practitioners report it as a real deployment bottleneck.

Isolating the heterogeneity source. Because cross-dataset heterogeneity bundles feature mismatch, fraud-rate shift (roughly two orders of magnitude between IEEE-CIS and PaySim), and domain shift (card-present vs. mobile-money) together, a controlled ablation isolates a single axis — e.g., same feature set, fraud rate artificially varied — so that any performance collapse can be attributed to a specific cause rather than reported as an unexplained effect.

Training. FedAvg, FedProx, SCAFFOLD, and a personalized baseline (Ditto or FedRep) are run on both the synthetic and real federations. Aggregation weighting (size-weighted vs. uniform-per-client) is treated as an explicit, reported design choice with a sensitivity check, since PaySim's transaction volume is roughly an order of magnitude larger than the other sources and would otherwise silently dominate a size-weighted average.

5. Evaluation Plan

- **Primary metrics:** AUC-PR and F1, per-client and aggregate (accuracy is uninformative under fraud's class imbalance).
- **Efficiency metric:** rounds to convergence, defined against a fixed AUC-PR plateau tolerance sustained over N consecutive rounds.
- **Headline deliverable:** a table/plot showing whether algorithm rankings change between the synthetic-split and real cross-institution settings, plus the ablation result attributing any flip to a specific heterogeneity axis.

6. Expected Contributions

1. First controlled comparison of FL algorithm rankings under synthetic vs. genuinely cross-institutional heterogeneity in fraud detection.
2. A documented, reusable schema-harmonization procedure for cross-institution fraud data.
3. An ablation methodology for attributing FL performance degradation to a specific heterogeneity source rather than reporting heterogeneity as a monolithic effect.

7. Pros and Cons

Pros

- Empirical/benchmarking format lowers execution risk relative to a novel-algorithm paper — the result is meaningful whichever way it comes out.
- All three datasets are public with no DUA or IRB process required, so the project can start immediately.
- Reuses an FL orchestration stack (Flower) already exercised in prior hands-on work, reducing tooling risk.
- Clear, falsifiable headline result: either the ranking flips (novel finding) or it doesn't (still a useful validation of current benchmarking practice).

Cons

- ULB's PCA-anonymized features limit the richest cross-dataset comparison to two datasets rather than three.

- PaySim is itself agent-simulated data, which partially undercuts the "real, not simulated" framing in the title and needs to be addressed directly in the paper.
- Heterogeneity sources (schema, fraud rate, domain) are naturally bundled together in a cross-dataset comparison and require deliberate ablation design to avoid an unattributed, inconclusive result.
- A roughly two-order-of-magnitude fraud base-rate gap between datasets could dominate results if not explicitly controlled for.

8. Resource Requirements and Practical Considerations

- **Compute:** Moderate. Tabular models (gradient-boosted trees or small MLPs/logistic models) run on a single GPU or even CPU — no large-model training required, making this the lowest-compute proposal of the three.
- **Team/skills needed:** FL orchestration experience (Flower), tabular feature engineering, familiarity with imbalanced-classification metrics.
- **Ethics/access:** None required beyond standard Kaggle terms of use — no patient data, no DUA.
- **Success criteria:** A clear, attributable answer to whether algorithm ranking changes between synthetic and real heterogeneity; the ablation should identify which specific heterogeneity axis (feature mismatch, fraud-rate shift, or domain shift) drives any observed change.
- **Risk mitigation:** If no ranking flip is observed, this is framed up front as still publishable — evidence that current synthetic benchmarking practice is more representative than commonly assumed is itself a useful, citable result.

9. Timeline

Weeks 1–2: schema harmonization and federation construction. Weeks 3–5: training runs across both federations and all four algorithms. Week 6: ablation study. Weeks 7–8: analysis and writeup.

Proposal 2: Vertical Federated Learning for Feature-Imbalanced Tabular Data

Abstract

Vertical Federated Learning (VFL) enables institutions holding the same patients (rows) but different feature sets (columns) to train collaboratively — a hospital with vitals and labs, another with clinical notes. Existing VFL work assumes each party's feature set is structurally complete. This proposal addresses the case where an entire feature modality is missing for a party outright, not merely sparse at the patient level, and introduces a federated split-autoencoder mechanism that imputes missing latent representations from correlated parties without exchanging raw data.

1. Problem Statement

Real institutional partners in a VFL network rarely hold complete modalities. This proposal treats **institution-level structural missingness** — an entire feature type absent for a whole party — as the primary object of study, distinct from ordinary per-patient missingness

(which is a well-studied imputation problem even outside FL, and is treated here only as a secondary condition).

2. Dataset Sources

Dataset	Source	Access	Role
MIMIC-III Clinical Database	PhysioNet — https://physionet.org/content/mimiciii/	Credentialed: requires completing the CITI "Data or Specimens Only Research" training course and signing a data use agreement; not instant, typically 1–2 weeks lead time	Primary structured-EHR source for both parties
MIMIC-IV	PhysioNet — https://physionet.org/content/mimiciv/	Same credentialing process as MIMIC-III	Alternative/updated structured-EHR source
MIMIC-CXR (optional extension)	PhysioNet — https://physionet.org/content/mimic-cxr/	Same PhysioNet credentialing, linked to MIMIC-IV subject IDs	Genuine imaging modality for the higher-effort cross-modality extension

Credentialing should be initiated before any other project work begins, since it is the longest lead-time item in the whole proposal and is not required at all for Proposals 1 or 3.

3. Data and Modality Design

The original plan (vitals vs. MRI on MIMIC-III) does not map onto the actual MIMIC-III release, which does not include MRI data. The design is revised as follows:

- **Primary setup:** two structured-modality parties within MIMIC-III/IV — Party A holds vitals and lab values, Party B holds medication and clinical-note-derived embeddings, for the same patient cohort. This is fully executable within MIMIC-III/IV alone.
- **Extension (optional, higher effort):** a genuine cross-modality pairing using MIMIC-CXR (chest X-ray) linked to MIMIC-IV structured data, giving a true imaging-vs-structured-data vertical split if the primary result motivates it.

Private set intersection (PSI) for sample alignment across parties is assumed to have already run; this is stated explicitly rather than left implicit, consistent with standard scoping in the VFL literature.

4. Proposed Methodology

Mechanism. A split-autoencoder learns a shared latent representation per party from whichever features are locally available. When a party's modality is entirely missing for the current training round, its latent contribution is imputed from the correlation structure learned across the other parties, and the global model dynamically weights available modalities rather than treating a missing one as zero-filled.

Missingness conditions, evaluated separately:

1. **Institution-level structural missingness (primary):** a party's modality is withheld entirely for a specified fraction of institutions/rounds.
2. **Random per-patient missingness (secondary/MCAR baseline):** individual patients lack a modality at random, for comparison against condition 1.
3. **Informative missingness (MNAR, added condition):** missingness is correlated with an outcome-relevant variable (e.g., sicker patients receive more lab tests), reflecting the fact that real clinical missingness is rarely random. This condition is necessary because a random-drop-only evaluation would test a substantially easier case than what institutions actually face.

Threat model. Split-learning intermediate activations are known to be vulnerable to reconstruction/model-inversion attacks. The privacy claim is evaluated empirically via a reconstruction-attack baseline against the shared latent representations, rather than asserting "no raw data exchanged" implies leakage-proof.

5. Evaluation Plan

- **Baselines:** zero-fill, mean/mode imputation, single-institution complete-case-only model, non-federated multimodal imputation network (upper-bound reference).
- **Primary metric:** predictive performance on patient mortality (MIMIC's standard outcome task) at 0%, 20%, and 40% structural feature missingness, under all three missingness conditions above.
- **Privacy metric:** reconstruction-attack success rate against shared latent representations.

6. Expected Contributions

1. A federated imputation mechanism for institution-level (not just per-patient) modality absence in VFL.
2. The first VFL evaluation, to this proposal's knowledge, distinguishing MCAR, institution-level structural, and MNAR missingness as separate conditions rather than one blended "missing data" setting.
3. An empirical reconstruction-attack evaluation of the privacy claim, rather than an architectural assertion alone.

7. Pros and Cons

Pros

- Addresses a genuinely underexplored gap — institution-level structural missingness in VFL — giving this the highest novelty ceiling of the three proposals.
- Revised modality design (vitals/labs vs. notes/medications) is fully executable within one credentialed dataset, without depending on the optional imaging extension.
- The MNAR condition captures a realistic clinical missingness pattern that most VFL papers ignore entirely.
- The reconstruction-attack evaluation gives the privacy claim actual empirical backing instead of resting on an architectural assertion.

Cons

- Requires PhysioNet credentialing, a non-trivial lead time and ethics-training requirement not shared by the other two proposals.
- Both simulated "institutions" are derived from a single MIMIC cohort, so genuine inter-institutional heterogeneity (different hospitals, different populations) is not actually captured.
- The MIMIC-CXR extension substantially raises complexity and scope if pursued.
- Split-autoencoder imputation under heavy institution-level missingness may hit a low performance ceiling that must be reported honestly rather than framed as a stronger result than it is.

8. Resource Requirements and Practical Considerations

- **Compute:** Autoencoder plus downstream classifier training — a single modern GPU is likely sufficient given MIMIC's tabular/text scale; this is not LLM-scale compute.
- **Ethics/access:** Patient data, de-identified but still governed by a PhysioNet data use agreement; no re-identification attempts are permitted under that agreement, and this constraint should be stated in any resulting paper.
- **Team/skills needed:** EHR data handling experience, autoencoder architectures, and — if the notes-embedding path is used — basic clinical NLP familiarity.
- **Success criteria:** the federated imputation approach should outperform zero-fill and mean-fill baselines under institution-level missingness at both 20% and 40%; the reconstruction-attack success rate should be reported honestly even if it is higher than hoped, rather than omitted or downplayed.
- **Risk mitigation:** if the primary structured-vs-structured setup shows limited signal, the MIMIC-CXR extension is scoped as an optional fallback to strengthen the modality contrast, not a required first step.

9. Timeline

Weeks 1–2: PhysioNet credentialing (start immediately, runs in parallel with early setup), data partitioning, and PSI assumption setup. Weeks 3–5: split-autoencoder implementation and training across missingness conditions. Week 6: reconstruction-attack evaluation. Weeks 7–8: analysis and writeup.

Proposal 3: Asynchronous Federated Unlearning for LLMs via Personalized LoRA Residuals

(Revised from "Cryptographic Sharding" — the original shard design forced a choice between two failure modes: private-per-client shards that aren't really federated, or shared-per-cluster shards where deleting one client's contribution over-forgets everyone else's. The design below resolves this with an explicit two-tier structure.)

Abstract

Federated fine-tuning of LLMs must comply with data deletion requests, but retraining a billion-parameter model per request is intractable. This proposal introduces a two-tier LoRA architecture that separates a collaboratively shared adapter from a private per-client residual adapter, so that a deletion request can remove exactly one client's contribution without destroying the collaborative benefit shared by the rest of the federation — and evaluates whether this actually satisfies unlearning under membership inference testing.

1. Problem Statement

A single global LoRA shard, shared and merged across many clients, cannot be selectively unlearned without degrading every other client sharing it. A fully private, per-client LoRA shard can be dropped trivially, but then no genuine collaborative learning has occurred — there is nothing "federated" to unlearn. This proposal treats that tension as the central design problem rather than a footnote.

2. Dataset and Model Sources

Resource	Source	Access	Role
WikiText-103	Hugging Face / originally Salesforce Research — https://huggingface.co/datasets/wikitext	Public, free, permissive license	General-knowledge partition, Tier 1 signal across all clients
Llama-3-8B or Mistral-7B-v0.3	Hugging Face	Public weights; Llama 3 Community License or Apache 2.0 (Mistral) — license terms should be checked for any research-use restrictions before publication	Frozen base model
Synthetic "forget set"	Constructed in-house	N/A — generated synthetic PII rather than sourced from real individuals, specifically to avoid using real people's data as the injected deletion target	Tier 2 signal, isolated per client

If a copyrighted excerpt is used instead of synthetic PII as a forget-set target, it must be used strictly for internal MIA testing and never reproduced in any paper, appendix, or shared artifact.

3. Proposed Architecture

Tier 1 — Shared adapter. A LoRA module aggregated across all clients via standard federated averaging with secure aggregation (clients' individual updates are not visible to the server in isolated form, only the aggregate). This tier captures genuinely general, non-client-identifying language patterns and is not deleted on a per-client request.

Tier 2 — Private residual adapter. Each client additionally trains a small private LoRA residual that captures client-specific patterns not captured by the shared tier. This residual is stored isolated per client (not merged into Tier 1) and is exactly what gets dropped on a deletion request.

Unlearning trigger. On a deletion request, the server drops the requesting client's Tier 2 residual immediately. Tier 1 is untouched, so no other client's contribution is degraded. This

resolves the original contradiction: collaborative benefit lives in Tier 1, and forgetting is scoped exactly to Tier 2, with no over-forgetting of other clients and no dependency on retraining as the unlearning mechanism itself.

Audit step. Because Tier 1 could in principle still memorize client-identifiable specifics through the aggregate, a periodic membership-inference audit checks whether Tier 1 alone leaks membership signal for any client above a set threshold. If it does, Tier 1 undergoes a scheduled, low-frequency retraining pass — reported honestly as a separate, non-instantaneous maintenance cost rather than folded into the "instantaneous" unlearning claim.

4. Experimental Setup

- **Frameworks:** Flower for federated orchestration, PyTorch and Hugging Face PEFT for LoRA.
- **Model:** Mistral-v0.3 or Llama-3 (7-8B), frozen base, both LoRA tiers trained on top.
- **Datasets:** WikiText partitioned across clients for general knowledge (Tier 1 signal); a distinct, injected per-client "forget set" (synthetic PII or a specific copyrighted excerpt) for Tier 2 signal, isolated to a single client each.

5. Evaluation Plan

- **Unlearning efficacy:** Membership Inference Attack (MIA) success rate against the forget set, both before and after Tier 2 is dropped — target is a drop to ~50% (random guessing).
- **Non-target-client integrity:** MIA and task performance for *other* clients, measured before and after a given client's deletion, to confirm Tier 1 isolation actually prevents collateral forgetting (this is the check the original single-shard design could not pass).
- **Model utility:** held-out perplexity, before and after deletion.
- **System efficiency:** wall-clock time for Tier 2 drop (expected to be near-instantaneous) reported separately from the wall-clock cost of a Tier 1 audit-triggered retrain (expected to be non-trivial), against full-model retraining as the baseline.

6. Expected Contributions

1. A two-tier LoRA architecture that gives a concrete, falsifiable answer to the shared-vs-private shard tradeoff in federated unlearning, rather than assuming it away.
2. An explicit, separately-reported cost model distinguishing instantaneous client-level forgetting from the rarer Tier 1 maintenance retrain.
3. An empirical test of collateral forgetting on non-target clients, which prior single-shard federated unlearning designs do not report.

7. Pros and Cons

Pros

- The two-tier design directly resolves the shard-granularity contradiction that undermines the original single-shard proposal, rather than leaving it unaddressed.

- Secure aggregation on Tier 1 gives genuine cryptographic substance to the design, unlike the original naming-only "cryptographic sharding."
- The collateral-forgetting evaluation on non-target clients is a check prior single-shard designs don't report, giving a concrete point of differentiation.
- MIA-based efficacy testing and perplexity-based utility testing are established, reviewer-expected methodologies, which reduces methodological risk.

Cons

- Federated unlearning is the most heavily published of the three subfields, raising the novelty bar this proposal has to clear.
- Requires the most compute of the three proposals by a wide margin (7-8B parameter models, even with a frozen base and LoRA).
- The audit-triggered Tier 1 retrain is still not fully instantaneous, and if audits trigger frequently in practice, the "fast unlearning" headline claim weakens.
- Synthetic forget-set injection may not fully represent how real client-data memorization manifests in practice at LoRA-adapter scale, which should be acknowledged as a limitation rather than assumed away.

8. Resource Requirements and Practical Considerations

- **Compute:** Significant relative to the other two proposals. A 7-8B parameter frozen base model needs at minimum a single high-memory GPU (e.g., A100 40GB/80GB class) for reasonable batch sizes even with LoRA; a cloud GPU budget should be planned for explicitly, unlike Proposals 1 and 2.
- **Copyright consideration:** if a copyrighted excerpt is used as an injected forget-set target, it must be used only internally for MIA testing and never reproduced in any paper, appendix, or artifact.
- **Team/skills needed:** distributed-systems experience for the secure aggregation implementation, plus LLM fine-tuning experience.
- **Success criteria:** MIA success on the forget set drops to ~50% immediately after Tier 2 removal; MIA success and perplexity for non-target clients remain statistically unchanged post-deletion; Tier 2 drop wall-clock time is orders of magnitude faster than full retraining.
- **Risk mitigation:** if compute budget is a binding constraint, a smaller open model (1-3B class) can substitute for the 7-8B target while preserving the same experimental design, at some cost to how convincingly the result generalizes to production-scale LLMs.

9. Timeline

Weeks 1-2: two-tier architecture and secure aggregation implementation. Weeks 3-4: training across simulated clients with injected forget sets. Week 5: MIA and collateral-forgetting evaluation. Week 6: audit/retrain cost measurement. Weeks 7-8: analysis and writeup.

Cross-Proposal Summary

	Proposal 1: Fraud Detection	Proposal 2: Vertical FL	Proposal 3: LLM Unlearning
Data access	Immediate (public, no DUA)	Delayed (PhysioNet credentialing, 1-2 weeks)	Immediate (public models/datasets)
Compute needs	Low	Moderate	High (7-8B parameter model)
Novelty ceiling	Moderate	Highest	Moderate (crowded subfield)
Execution risk	Lowest	Moderate	Highest
Ethics/IRB overhead	None	Yes (patient data DUA)	None